CORAL run 2026-10-09_162748 — results (single seed: S3-FIFO)

Run summary

Item	Value
Task	cache-ext-get-scan (LevelDB GET/SCAN, cache_ext eviction-policy search)
Setup	Branch 20-evol-one-seed: S3-FIFO is the ONLY seed (seed/noop.c holds it; no other example policies are given to the agent). Score = measured throughput in ops/s (absolute; higher is better), no baseline calibration.
Run start → stop (local)	16:27:48 → 17:31:01 (63.2 min)
Stop reason	max_real_attempts (cap 20, real attempts 20); triggering attempt 5638aedb, score 1232.54
Agent	1 agent (captain-nemo), runtime claude_code, model alias claude-opus-5-5 → Bedrock us.anthropic.claude-opus-5-5
Attempts recorded	29 = 20 real + 9 --tune (tune attempts are graded identically but do not count toward the cap); 2 of them crashed without a score (#8, #13); coral eval invocations: 31
Attempts by code	2 ran the unmodified S3-FIFO seed (#1, #29); 25 scored agent-evolved variants
Attempt status (CORAL)	improved: 3, regressed: 24, crashed: 2 (relative to the agent's own previous best)
Agent sessions (claude -p)	31; terminal reason: aborted_streaming 26, aborted_tools 3, completed 2
LLM usage (agent)	83,788 output tokens; 8,824,423 cache-read + 260,950 cache-write + 186 input tokens; Claude Code cost estimate $4.95; API time 1,945 s
Notes / skills written	0 notes saved (3 agent actions were denied by the permission layer); skills present: create-notes, deep-research, organize-files, skill-creator

Headline numbers

Metric	Value	Metric	Value
Unmodified S3-FIFO seed, this run	1,252.27, 1,232.54 ops/s (mean 1,242.4)	Seed, all 3 runs pooled (n=5)	1,246.4 ± 18.0 ops/s (range 1,222.5 … 1,263.0)
Highest score (any attempt)	#24 260db443: 1,276.89 ops/s (+2.78% vs seed mean; tune/instrumented)	Best real evolved attempt	#5 30ecfc5d: 1,276.64 ops/s (+2.76% vs seed mean)
Re-run of that best real design	#5: 1,276.64 / #7: 1,223.61	Evolved attempts above seed mean	8 / 25 (above the pooled seed max: 3)
Throughput, mean ± sd of 27 scored	1,224.1 ± 35.8 ops/s	Worst score	#9: 1,097.61 ops/s
Final attempt (#29)	seed re-run: 1,232.54 ops/s	Reference: earlier noop baseline (run 1, external)	1,224.3 ± 9.5 ops/s (n=8)

The reference for every Δ in this report is the mean of the unmodified S3-FIFO seed measured in this run. Single evals of identical code vary by roughly 1–4% here, so a Δ of that size is not evidence of improvement; the noise section quantifies it.

Scoring and recorded probes

Probe	Source	Role in this run	Unit
throughput	results.json: throughput_ops_per_sec	THE SCORE (maximized)	ops/s
rbytes	cgroup io.stat (delta over measured phase)	diagnostic only	MiB
wbytes, rios	cgroup io.stat	diagnostic only	MiB, count
refault_file, pgmajfault, pgactivate, resident_file	cgroup memory.stat	diagnostic only	count, count, count, MiB
wallclock	benchmark wall time (setup + warm-up + measured)	diagnostic only	s
policy_counters	metrics file the agent's own policy exports (from attempt #13 on)	diagnostic only, agent-defined	count

Environment and workload

Evaluation node (CloudLab, Wisconsin)

Component	Value	Component	Value
Node type	c220g1	Kernel	6.6.8-cache-ext+ (Ubuntu 22.04.2)
CPU	2 × Intel Xeon E5-2630 v3 @ 2.40 GHz	Cores / threads	16 cores / 32 threads (2 sockets, 2 NUMA nodes)
Memory	131,732,952 kB (125.6 GiB), no swap	Disks	2 × 1.1 TB SAS HDD, 1 × 447 GB SSD
Benchmark filesystem	/mydata on /dev/sda3 (HDD), 63 GB	Cgroup	cgroup v2; benchmark memory.max = 64 MB
Compiler / tools	clang 14.0.0-1ubuntu1.1, bpftool v7.3.0	Eval isolation	one eval at a time (flock); policy attached via BPF struct_ops

Workload parameters (benchmark harness defaults, every attempt)

Parameter	Value	Parameter	Value
Benchmark	My-YCSB (leveldb-scan branch), run_leveldb	Request distribution / Zipfian constant	zipfian / 0.99
Operation mix	95% GET (read), 5% SCAN	Worker threads	4 (exactly 1 dedicated scan thread)
Scan length	500 keys per SCAN	Measured phase	30 s
Database entries	500,000	Warm-up phase	10 s (same mix, probes not recording)
Key / value size	16 B / 200 B	LevelDB block cache	16 MiB
Cgroup memory limit (page cache)	64 MB	Page cache at start	dropped before each run
Per-eval timeout	180 s	Observed eval wall time	39.7 s mean (39.5 … 40.0)

Per-attempt results

#	Local	Commit	Parent	Policy	Class / status	SCORE = throughput (ops/s)	Δ ops/s vs seed mean	Δ%	Wall s	Eval s	+/− lines	Code lines
1	16:28:36	8f915176	bb1f29ba	S3-FIFO seed (unmodified)	real / improved	1,252.27	+9.9	+0.79	39.8	45	+1/−0	603
2	16:32:33	1a501cb5	8f915176	evolved	real / regressed	1,250.00	+7.6	+0.61	39.5	46	+98/−6	695
3	16:34:37	206568d5	1a501cb5	evolved	real / regressed	1,237.15	-5.3	-0.42	40.0	46	+3/−3	695
4	16:36:16	59fd1ebc	206568d5	evolved	real / regressed	1,233.92	-8.5	-0.68	39.7	46	+4/−4	695
5	16:38:21	30ecfc5d	59fd1ebc	evolved	real / improved	1,276.64	+34.2	+2.76	39.6	46	+20/−16	699
6	16:39:35	a2448356	30ecfc5d	evolved	real / regressed	1,254.07	+11.7	+0.94	40.0	46	+1/−1	699
7	16:40:45	b0ac1a55	30ecfc5d	evolved	real / regressed	1,223.61	-18.8	-1.51	39.5	46	+1/−0	700
9	16:42:28	9ed2e60b	7aae703e	evolved	real / regressed	1,097.61	-144.8	-11.65	39.5	46	+2/−0	701
10	16:44:10	c13ebbf7	9ed2e60b	evolved	real / regressed	1,217.87	-24.5	-1.97	39.6	46	+3/−5	699
11	16:46:23	3f206358	c13ebbf7	evolved	tune / regressed	1,263.37	+21.0	+1.69	39.8	46	+1/−0	700
12	16:47:56	1713bf62	3f206358	evolved	real / regressed	1,207.87	-34.5	-2.78	39.7	46	+5/−4	701
14	16:50:24	04e109e4	4022170b	evolved	tune / regressed	1,254.33	+11.9	+0.96	39.8	43	+3/−3	733
15	16:51:54	f368d683	04e109e4	evolved	tune / regressed	1,262.25	+19.8	+1.60	39.7	46	+48/−15	766
16	16:54:07	151b38d5	f368d683	evolved	tune / regressed	1,242.68	+0.3	+0.02	39.6	46	+14/−4	776
17	16:55:34	1a65212c	151b38d5	evolved	real / regressed	1,212.20	-30.2	-2.43	39.6	46	+9/−1	784
18	16:57:22	e553fcdd	1a65212c	evolved	real / regressed	1,211.25	-31.2	-2.51	40.0	46	+1/−1	784
19	16:58:47	2c4c9038	e553fcdd	evolved	real / regressed	1,192.40	-50.0	-4.02	39.6	46	+11/−5	790
20	17:02:31	864bdf61	151b38d5	evolved	tune / regressed	1,195.18	-47.2	-3.80	39.7	46	+19/−0	795
21	17:06:14	17dee3fd	864bdf61	evolved	real / regressed	1,211.39	-31.0	-2.50	39.5	46	+11/−12	794
22	17:07:52	7926ed02	17dee3fd	evolved	real / regressed	1,198.43	-44.0	-3.54	39.7	46	+4/−4	794
23	17:10:13	6843b181	864bdf61	evolved	tune / regressed	1,207.71	-34.7	-2.79	39.6	46	+39/−2	832
24	17:14:23	260db443	6843b181	evolved	tune / improved	1,276.89	+34.5	+2.78	39.6	46	+19/−10	841
25	17:18:51	9153cad4	260db443	evolved	tune / regressed	1,191.78	-50.6	-4.07	39.5	46	+12/−2	851
26	17:25:02	7e4693b2	9153cad4	evolved	real / regressed	1,222.98	-19.4	-1.56	39.6	46	+8/−0	859
27	17:26:57	2373b36f	7e4693b2	evolved	real / regressed	1,213.84	-28.6	-2.30	39.7	46	+0/−10	849
28	17:28:48	58cf40cf	2373b36f	evolved	real / regressed	1,211.07	-31.3	-2.52	39.6	46	+1/−1	849
29	17:30:15	5638aedb	8f915176	S3-FIFO seed (unmodified)	real / regressed	1,232.54	-9.9	-0.79	39.9	46	+1/−1	603

Shaded = highest score. Δ columns compare with the mean of the unmodified seed in this run (1,242.4 ops/s). Class: real = counts toward the cap, tune = graded the same but not counted. Eval s = submit → graded. Status is CORAL's comparison with the agent's own previous best, not with a baseline.

Diagnostic probes and derived per-operation metrics

#	Commit	Policy	Bytes read MiB	Δ% vs seed mean	Bytes written MiB	Read I/Os	Refaults	Δ% vs seed mean	pgmajfault	pgactivate	Resident file MiB	Ops in phase (≈ tput × 30)	Read KB / op	Refaults / 1k ops	Read I/Os / 1k ops	Read MB/s
1	8f915176	S3-FIFO seed (unmodified)	314.6	+1.4	0.0	22,719	53,786	+2.1	31	14,946	47	37,568	8.78	1,431.7	604.7	11.00
2	1a501cb5	evolved	296.7	-4.4	0.0	24,005	49,380	-6.3	16	11,971	45	37,500	8.30	1,316.8	640.1	10.37
3	206568d5	evolved	310.2	-0.0	0.0	22,533	52,672	-0.0	31	15,196	47	37,114	8.76	1,419.2	607.1	10.84
4	59fd1ebc	evolved	297.6	-4.1	0.0	23,779	49,584	-5.9	16	12,961	45	37,018	8.43	1,339.5	642.4	10.40
5	30ecfc5d	evolved	293.1	-5.5	0.0	24,172	48,444	-8.1	21	12,870	46	38,299	8.02	1,264.9	631.1	10.24
6	a2448356	evolved	287.2	-7.4	0.0	23,668	46,924	-10.9	19	12,483	46	37,622	8.00	1,247.2	629.1	10.04
7	b0ac1a55	evolved	295.5	-4.8	0.0	23,617	49,018	-7.0	16	13,005	45	36,708	8.44	1,335.3	643.4	10.33
9	9ed2e60b	evolved	273.6	-11.8	0.0	21,559	43,511	-17.4	16	12,071	45	32,928	8.71	1,321.4	654.7	9.56
10	c13ebbf7	evolved	296.8	-4.3	0.0	23,807	49,316	-6.4	16	13,179	46	36,536	8.52	1,349.8	651.6	10.37

#	Commit	Policy	Bytes read MiB	Δ% vs seed mean	Bytes written MiB	Read I/Os	Refaults	Δ% vs seed mean	pgmajfault	pgactivate	Resident file MiB	Ops in phase (≈ tput × 30)	Read KB / op	Refaults / 1k ops	Read I/Os / 1k ops	Read MB/s
11	3f206358	evolved	295.7	-4.7	0.0	24,201	49,111	-6.8	22	13,790	46	37,901	8.18	1,295.8	638.5	10.34
12	1713bf62	evolved	294.0	-5.2	0.0	21,810	48,604	-7.7	31	13,374	47	36,236	8.51	1,341.3	601.9	10.28
14	04e109e4	evolved	297.2	-4.2	0.0	24,027	49,447	-6.1	16	13,578	46	37,630	8.28	1,314.0	638.5	10.39
15	f368d683	evolved	289.2	-6.8	0.0	23,716	47,381	-10.1	28	13,918	46	37,868	8.01	1,251.2	626.3	10.11
16	151b38d5	evolved	297.8	-4.0	0.0	23,978	49,577	-5.9	16	13,147	46	37,280	8.38	1,329.8	643.2	10.41
17	1a65212c	evolved	285.6	-7.9	0.0	23,138	46,547	-11.7	16	12,644	45	36,366	8.23	1,280.0	636.3	9.98
18	e553fcdd	evolved	289.0	-6.8	0.0	22,796	47,371	-10.1	24	13,358	46	36,338	8.34	1,303.6	627.3	10.10
19	2c4c9038	evolved	296.6	-4.4	0.0	23,599	49,350	-6.3	16	12,860	46	35,772	8.69	1,379.6	659.7	10.37
20	864bdf61	evolved	301.8	-2.7	0.0	21,963	50,582	-4.0	31	14,074	47	35,855	8.83	1,410.7	612.5	10.55
21	17dee3fd	evolved	292.8	-5.6	0.0	23,639	48,340	-8.2	16	13,113	45	36,342	8.45	1,330.2	650.5	10.23
22	7926ed02	evolved	288.4	-7.0	0.0	22,521	47,235	-10.3	31	12,071	47	35,953	8.41	1,313.8	626.4	10.08
23	6843b181	evolved	291.4	-6.1	0.0	23,267	48,039	-8.8	16	12,948	46	36,231	8.43	1,325.9	642.2	10.19
24	260db443	evolved	292.1	-5.9	0.0	24,126	48,207	-8.5	21	12,706	46	38,307	8.00	1,258.4	629.8	10.21
25	9153cad4	evolved	296.6	-4.4	0.0	23,467	49,308	-6.4	16	12,743	45	35,753	8.70	1,379.1	656.4	10.37
26	7e4693b2	evolved	290.9	-6.2	0.0	23,316	47,857	-9.2	21	13,816	46	36,689	8.31	1,304.4	635.5	10.17
27	2373b36f	evolved	300.6	-3.1	0.0	22,002	50,257	-4.6	31	13,958	47	36,415	8.66	1,380.1	604.2	10.51
28	58cf40cf	evolved	287.8	-7.2	0.0	22,995	47,016	-10.8	20	13,329	46	36,332	8.31	1,294.1	632.9	10.06
29	5638aedb	S3-FIFO seed (unmodified)	305.9	-1.4	0.0	21,951	51,586	-2.1	31	14,281	47	36,976	8.67	1,395.1	593.7	10.69

Reference = unmodified seed mean in this run: bytes read 310.2 MiB, refaults 52,686. Assumes the probes cover exactly the 30 s measured phase, so operations = throughput × 30.

Attempts without a score (benchmark did not produce a result)

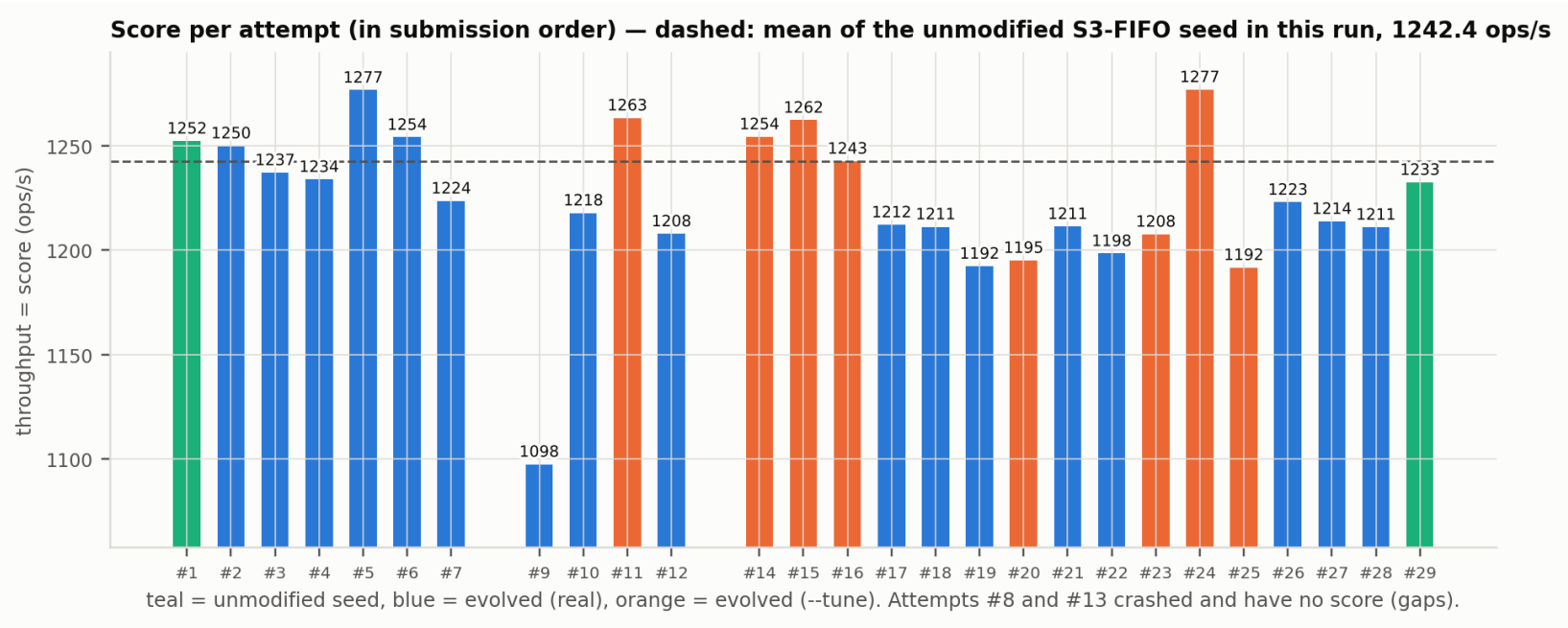
#	Commit	Class	Wall s	First line of grader feedback
8	7aae703e	real	3.1	eval: [FAIL] wall=3.12s error=benchmark[all] returned 1
13	4022170b	tune	-	[--tune mode] This grader does not differentiate tune mode from a real submission: scoring runs the full evaluation either way and returns the same sc

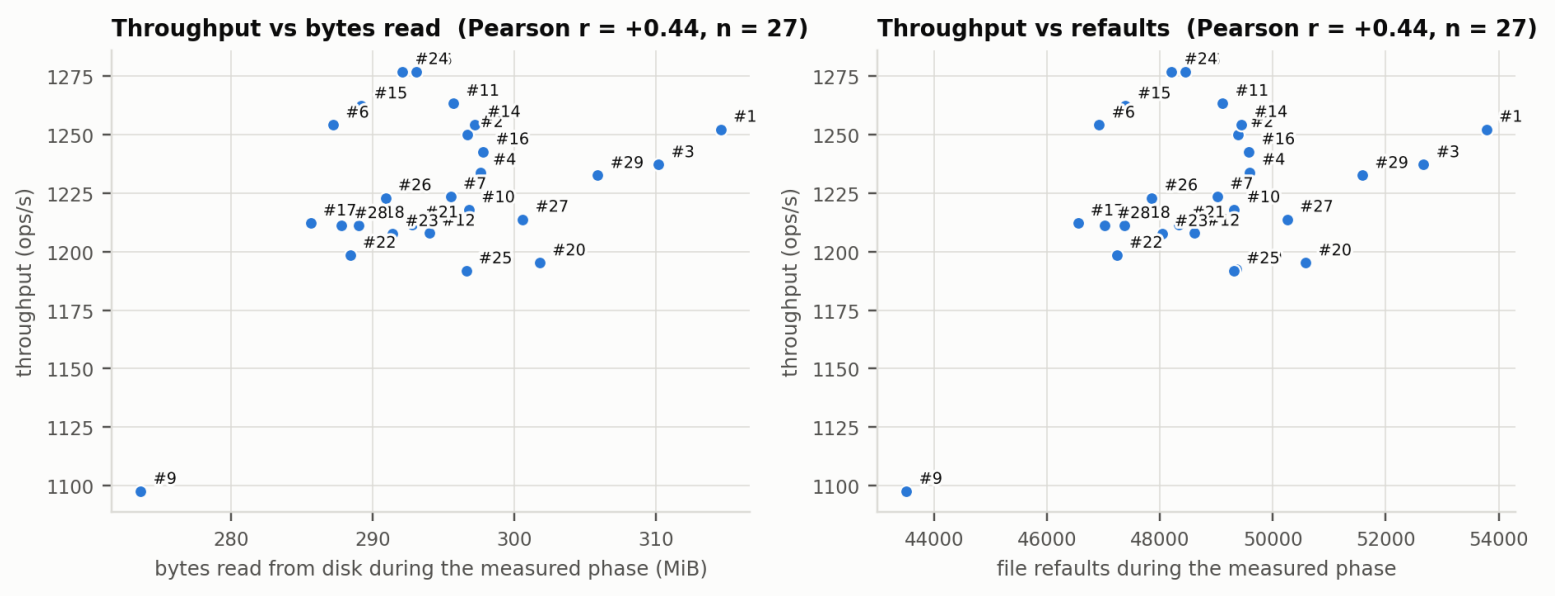
Attempt descriptions (commit messages, verbatim)

#	Commit	Class	Description
1	8f915176	real	baseline: unchanged S3-FIFO seed (comment-only change)
2	1a501cb5	real	scan-list v1: per-(tid,inode) sequential-insert detector (run>=3, gap<=64) routes scan folios to a separate FIFO scan_list evicted first (keep >=512 resident for readahead); scan folios skip ghost map. Rest is S3-FIFO.
3	206568d5	real	freq cap 3->7 in folio_accessed (Zipf: separate very-hot from warm in main CLOCK); on top of scan-list v1
4	59fd1ebc	real	revert freq cap to 3; main eviction via bpf_cache_ext_list_sample (sample 10, score=freq--, lowest evicted) instead of up-to-4-pass CLOCK iterate, to test whether eviction CPU cost / LFU-sampling helps
5	30ecfc5d	real	LFU-with-history: ghost map stores u32 access count (4x cache entries), restored on ghost-hit readmission; main evicted by sampling 32 and taking lowest count (no decrement/aging); uncapped freq. Small FIFO + scan list unchanged.
6	a2448356	real	LFU-with-history: ghost/count history 4x -> 16x cache_size entries (keep counts for more of the Zipf tail)
7	b0ac1a55	real	re-run of 30ecfc5 (LFU-with-history, 4x ghost) to check whether 1276.6 is real or noise
8	7aae703e	real (crashed)	DIAGNOSTIC: LFU-hist (30ecfc5) with folio_accessed hook removed from ops struct -> no per-hit BPF cost (degenerates to FIFO+ghost). Tests whether throughput is CPU/hook-bound rather than miss-bound; identical-code rerun showed 4% noise (1276.6 vs 1223.6).
9	9ed2e60b	real	DIAGNOSTIC retry: folio_accessed compiled out with #if 0 (unreferenced struct_ops prog fails load with attach_btf_id 0). Tests CPU/hook-bound vs miss-bound.
10	c13ebbf7	real	LFU-hist + 16x history (eval6 cfg) + SCAN_MIN 512->128 pages (give scan reserve back to GET set). Hook restored.
11	3f206358	tune	tune probe: same as c13ebbf (check what tune mode does)
12	1713bf62	real	structural attempt 1/3 (scan-sensitivity lane) DIAGNOSTIC: scan-detected folios inserted into main with count 8 (protected) instead of evict-first scan list. Big up => scan thread bounds throughput; big down => GET hit rate bounds it.
13	4022170b	tune (crashed)	structural attempt 2/3 INSTRUMENTATION: scan-list restored (=c13ebbf policy) + bss counters dumped by loader to stderr at SIGTERM with exit 1 so they surface in feedback

#	Commit	Class	Description
14	04e109e4	tune	structural attempt 2/3 INSTRUMENTATION retry: counters as unsigned long long (skel.h has no u64 in userspace)
15	f368d683	tune	structural attempt 2/3 INSTRUMENTATION v3: loader writes counters JSON every 1s to ./policy_metrics(.json) and any *METRIC* env path (guessing the policy_metrics convention)
16	151b38d5	tune	INSTRUMENTATION v4: + scan-folio accesses, evicted-with-freq0 per class, current list sizes
17	1a65212c	real	structural attempt 3/3 scan lane: scan list second-chance — evict consumed scan folios (freq>0) immediately, unconsumed readahead gets one CLOCK pass. Counters (eval16) showed 21477/41135 scan evictions were never read (~84MB wasted IO).
18	e553fcdd	real	scan lane follow-up: SCAN_MIN 128 -> 6000 (scan FIFO holds ~37% of cache) to catch repeat scans of Zipf-hot ranges; counters show GET set barely evicted (main 0, small ~3k) so cache is available. Keeps second-chance.
19	2c4c9038	real	fill eviction requests: chain scan->small/main->other->scan-below-min until request_nr satisfied (eval16 counters: got 13.0k vs req 24.0k => shortfalls => reclaim retries). Second-chance reverted, SCAN_MIN 128.
20	864bdf61	tune	INSTRUMENTATION v5: policy = eval16 (LFU-hist + scan list, SCAN_MIN 128); counters snapshot at t=12s, report steady-state deltas (d_*)
21	17dee3fd	real	structural attempt 1/3 'full-cache CLOCK for GET folios': all non-scan folios -> main (no small one-hit filter), main evicted by seed's CLOCK iterate (freq cap 3, ghost count restores freq), scan list still evicted first. Rationale: steady-state counters show page cache sees LevelDB block-cache misses (20k non-scan adds, 8.3k accesses, 4.6k small evictions, 2.4k ghost hits) => long reuse distances.
22	7926ed02	real	structural attempt 2/3 'full-cache residency for GET folios': same routing (all non-scan -> main, scan evicted first) but main evicted by LFU-hist sampling (32 samples, uncapped counts restored from ghost) instead of CLOCK
23	6843b181	tune	counter-drift fix attempt: fentry/delete_from_page_cache_batch drops metadata + list-size accounting for truncated folios (theory: compaction/file deletion removes folios without folio_evicted => small_size inflated => S3-FIFO balance logic broken, main never evicted). Policy otherwise = eval16. Reports trunc count.
24	260db443	tune	INSTRUMENTATION v6: why eviction requests go unfilled (got~req/2): count iterate outcomes (scan/small notready, evict, promote) + short calls. trunc theory refuted (trunc=2).
25	9153cad4	tune	INSTRUMENTATION v7: split scan-list not-ready visits (3.66M in eval24!) by cause: !uptodate / !lru / dirty-wb, and whether metadata is missing (stale node)
26	7e4693b2	real	structural attempt 1/3 'age-gated scan eviction': scan folios evicted only if consumed (freq>0) or unconsumed for >200ms. v7 counters: scan list is mostly in-flight readahead (1.43M !uptodate visits); folios were evicted the instant IO completed, before the scan read them (55% evicted unused).
27	2373b36f	real	structural attempt 2/3 age-gated scan eviction: same policy as 7e4693b, hot-path per-node diagnostic atomics removed (clean final candidate). Policy: per-(tid,inode) sequential scan detector -> scan list evicted first (consumed or >200ms unconsumed); S3-FIFO small + LFU-with-history main (ghost stores counts, 4x cache).
28	58cf40cf	real	structural attempt 3/3 age-gated scan eviction: scan iterate evict_list=small_list so kernel-rejected scan candidates (mode-B runs: 78k proposed, 4.8k evicted, 2600 short calls) migrate to S3-FIFO small queue instead of being re-proposed forever
29	5638aedb	real	seed S3-FIFO re-run #2 (noise baseline: is anything actually better than the seed?)

Score per attempt

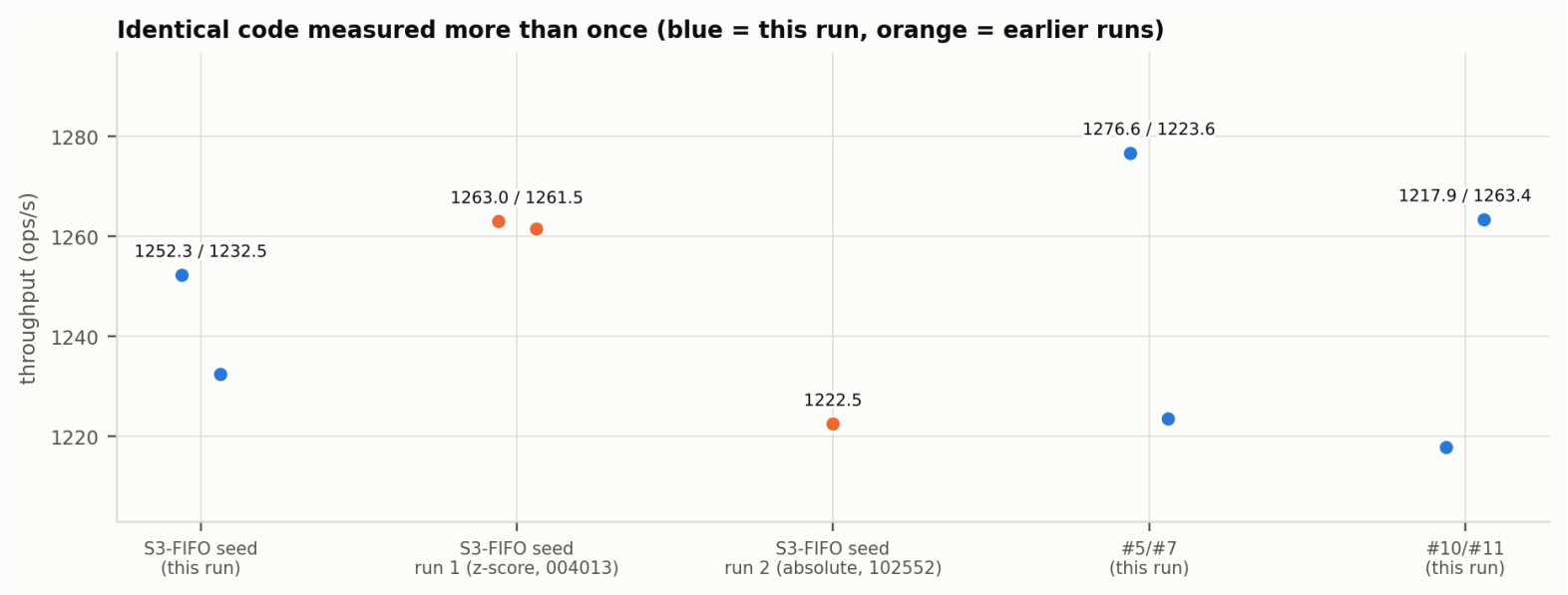




Is anything better than the seed? Repeated measurements of identical code

Code	Measurements (ops/s)	n	Mean	sd	Range % of mean
S3-FIFO seed, this run (#1, #29)	1,252.27, 1,232.54	2	1,242.4	14.0	1.59
S3-FIFO seed, run 1 (z-score, 004013)	1,262.97, 1,261.52	2	1,262.2	1.0	0.11
S3-FIFO seed, run 2 (absolute, 102552)	1,222.54	1	1,222.5	-	0.00
S3-FIFO seed, all runs pooled	1,252.3, 1,232.5, 1,263.0, 1,261.5, 1,222.5	5	1,246.4	18.0	3.24
Evolved, attempts #5/#7: LFU-with-history: ghost map stores u32 access count (4x cach	1,276.64, 1,223.61	2	1,250.1	37.5	4.24
Evolved, attempts #10/#11: LFU-hist + 16x history (eval6 cfg) + SCAN_MIN 512->128 pages	1,217.87, 1,263.37	2	1,240.6	32.2	3.67

Identical (comment-stripped) code is grouped by hash. The best real evolved design (#5, 1,276.64 ops/s) was re-run as #7 (1,223.61 ops/s); the seed itself spans 1,222.5 … 1,263.0 across five measurements in three runs. Differences between attempts of a few percent are therefore within measurement noise, and no evolved design is separated from the seed by more than that spread.



Best attempt and best agent-evolved design

Item	Highest score (overall)	Best agent-evolved design
Attempt	#24 260db443b4564b5c547829bf1e2097edeafbffb5	#5 30ecfc5db23287e26623475d1c1212760e1c72a8
Policy	evolved [tune]	evolved (best real, non-tune evolved attempt)
Score (= throughput, ops/s)	1,276.89	1,276.64
vs seed mean (this run)	+34.5 ops/s (+2.78%)	+34.2 ops/s (+2.76%)
Bytes read (MiB) / refaults	292.1 / 48,207	293.1 / 48,444
Code size	841 lines (kernel 556, loader 285)	699 lines (kernel 487, loader 212)
Description	INSTRUMENTATION v6: why eviction requests go unfilled (got~req/2): count iterate outcomes (scan/small notready, evict, promote) + short calls. trunc theory refuted (trunc=2).	LFU-with-history: ghost map stores u32 access count (4x cache entries), restored on ghost-hit readmission; main evicted by sampling 32 and taking lowest count (no decrement/aging); uncapped freq. Small FIFO + scan list unchanged.

Ranking of all attempts by score

Rank	#	Commit	Policy	Score (ops/s)	Δ vs best (ops/s)	Δ vs best (%)
1	24	260db443	evolved	1,276.89	+0.0	+0.00
2	5	30ecfc5d	evolved	1,276.64	-0.2	-0.02
3	11	3f206358	evolved	1,263.37	-13.5	-1.06
4	15	f368d683	evolved	1,262.25	-14.6	-1.15
5	14	04e109e4	evolved	1,254.33	-22.6	-1.77
6	6	a2448356	evolved	1,254.07	-22.8	-1.79
7	1	8f915176	S3-FIFO seed (unmodified)	1,252.27	-24.6	-1.93
8	2	1a501cb5	evolved	1,250.00	-26.9	-2.11
9	16	151b38d5	evolved	1,242.68	-34.2	-2.68
10	3	206568d5	evolved	1,237.15	-39.7	-3.11
11	4	59fd1ebc	evolved	1,233.92	-43.0	-3.37
12	29	5638aedb	S3-FIFO seed (unmodified)	1,232.54	-44.4	-3.47
13	7	b0ac1a55	evolved	1,223.61	-53.3	-4.17
14	26	7e4693b2	evolved	1,222.98	-53.9	-4.22
15	10	c13ebbf7	evolved	1,217.87	-59.0	-4.62
16	27	2373b36f	evolved	1,213.84	-63.1	-4.94
17	17	1a65212c	evolved	1,212.20	-64.7	-5.07
18	21	17dee3fd	evolved	1,211.39	-65.5	-5.13
19	18	e553fcdd	evolved	1,211.25	-65.6	-5.14
20	28	58cf40cf	evolved	1,211.07	-65.8	-5.15
21	12	1713bf62	evolved	1,207.87	-69.0	-5.41
22	23	6843b181	evolved	1,207.71	-69.2	-5.42
23	22	7926ed02	evolved	1,198.43	-78.5	-6.14
24	20	864bdf61	evolved	1,195.18	-81.7	-6.40
25	19	2c4c9038	evolved	1,192.40	-84.5	-6.62
26	25	9153cad4	evolved	1,191.78	-85.1	-6.67
27	9	9ed2e60b	evolved	1,097.61	-179.3	-14.04

Each attempt is a single measurement; the unmodified seed alone spanned 3.2% across five measurements, so adjacent ranks (differences under about 1–2%) are not distinguishable. The agent-evolved design's source is saved as best_evolved_policy.c.

Timing

Event	Time	Elapsed since start (min)
Run directory created / manager started	16:27:48	0.0
First model request at the gateway	16:27:55	0.1
First submission (attempt #1)	16:28:36	0.8
Last submission (attempt #29)	17:30:15	62.5
Last attempt graded / auto-stop recorded	17:31:01	63.2

Time budget	Seconds	% of run wall time
Evaluation (all 29 attempts, submit → graded)	1,235	32.6
Everything else (agent reading, thinking, editing)	2,558	67.4
Run wall time	**3,793**	**100.0**

Eval latency per attempt: min 42.8 s, median 45.9 s, max 46.3 s.

Agent sessions and LLM usage

#	Start	Wall s	API s (Δ)	Turns	Cost $ (Δ)	Input tok	Cache write tok	Cache read tok	Output tok	Thinking tok	API retries	Denials	Terminal reason
0	16:27:54	90	60	9	0.451	14	66,377	348,163	2,479	1,554	0	0	aborted_streaming
1	16:29:24	235	167	8	0.346	14	15,768	445,999	8,871	3,842	0	2	aborted_streaming
2	16:33:19	125	74	6	0.174	10	6,804	349,319	3,490	2,673	0	0	aborted_streaming
3	16:35:25	100	46	3	0.057	4	2,466	91,037	1,301	728	0	0	aborted_streaming
4	16:37:05	125	62	5	0.170	8	7,512	289,731	3,745	1,991	0	0	aborted_streaming
5	16:39:10	75	22	3	0.040	4	2,133	101,015	463	114	0	0	aborted_streaming
6	16:40:25	70	13	3	0.040	4	1,529	103,824	563	225	0	0	aborted_streaming
7	16:41:36	40	22	3	0.058	4	2,412	105,353	1,248	707	0	0	aborted_streaming
8	16:42:16	60	9	3	0.051	4	3,512	107,765	593	91	0	0	aborted_streaming
9	16:43:16	100	50	3	0.098	4	2,578	111,277	3,152	2,369	0	0	aborted_tools
10	16:44:56	23	23	1	0.093	4	8,364	113,855	1,406	455	0	0	completed
11	16:45:21	110	55	5	0.158	8	5,645	372,105	2,782	2,334	0	0	aborted_streaming
12	16:47:11	95	18	3	0.081	4	6,299	127,864	1,179	467	0	0	aborted_streaming
13	16:48:47	85	61	4	0.164	6	7,997	273,188	3,446	808	0	0	aborted_streaming
14	16:50:12	60	9	3	0.062	4	5,086	142,160	393	0	0	0	aborted_streaming
15	16:51:12	90	33	3	0.097	4	4,669	147,246	2,198	690	0	0	aborted_streaming
16	16:52:42	135	70	4	0.195	6	10,051	309,896	4,132	2,510	0	0	aborted_streaming
17	16:54:58	85	32	3	0.095	4	4,273	161,966	2,064	921	0	0	aborted_streaming
18	16:56:23	110	53	3	0.129	4	5,996	166,239	3,268	2,650	0	0	aborted_streaming
19	16:58:13	80	30	3	0.105	4	7,223	172,235	1,701	577	0	0	aborted_tools
20	16:59:33	16	16	1	0.081	4	5,417	179,458	888	141	0	0	completed
21	16:59:53	205	137	8	0.604	14	10,140	1,129,822	6,160	3,748	0	0	aborted_streaming
22	17:03:19	225	171	3	0.153	4	6,197	195,015	4,137	2,670	0	0	aborted_streaming
23	17:07:04	95	43	3	0.126	4	8,326	201,212	2,201	1,329	0	0	aborted_streaming
24	17:08:39	140	80	4	0.247	6	10,654	425,354	5,413	3,569	0	0	aborted_streaming
25	17:10:59	250	184	3	0.145	4	5,458	220,192	3,694	1,448	0	0	aborted_streaming
26	17:15:10	270	188	3	0.232	8	8,072	459,372	4,983	3,567	0	0	aborted_streaming
27	17:19:40	370	110	6	0.317	10	12,351	967,220	3,081	926	0	1	aborted_streaming
28	17:25:50	115	41	4	0.174	6	6,622	496,628	2,096	931	0	0	aborted_streaming
29	17:27:46	110	47	3	0.113	4	4,523	252,695	1,990	1,264	0	0	aborted_streaming
30	17:29:36	85	21	3	0.097	4	6,496	257,218	671	314	0	0	aborted_tools
Σ		**3,767**	**1,945**	**119**	**4.95**	**186**	**260,950**	**8,824,423**	**83,788**	**45,613**	**0**	**3**	

Cost and API time are cumulative in Claude Code's result records, so rows show the increase over the previous session and Σ the final value; cost is Claude Code's estimate, not an AWS bill. Wall s, turns and tokens are per session. aborted_streaming = the session was cut off mid-response (consistent with CORAL interrupting and resuming the agent).

Agent actions

Tool	Calls
Bash	55
WebFetch	2
Write	1
WebSearch	1

coral command	Count
eval	22
eval --tune	9
checkout	4
log	3
wait	1
diff	1

Permission denials (actions the permission layer refused)

Session	Tool	Target (truncated)
1	Bash	{"command": "mkdir -p .claude/notes/experiments && python .claude/skills/create-notes/scripts/stamp.py experiment --out .claude/notes/experiments/eval-1-baselin
1	Write	{"file_path": "/home/aanil/Desktop/ldos/arjua-forks/coral-cache-evolution/results/cache-ext-get-scan/2026-10-09_162748/agents/captain-nemo/.claude/notes/experim
27	Bash	{"command": "python3 - <<'EOF'\np='noop.c'; s=open(p).read()\ns=s.replace(\"\"\"\tbool in_main;\n\tbool scan;\n};\"\"\",\"\"\"\tbool in_main;\n\tbool scan;\n\tu

LLM gateway (LiteLLM → Bedrock)

Requests by model and HTTP status

Model alias	200	400	503	Total	Success %
claude-opus-5-5	133	33	5	171	77.8
claude-sonnet-5	0	29	0	29	0.0
all	**133**	**62**	**5**	**200**	**66.5**

Failure causes

HTTP	Model alias	Count	Message (summarised from the gateway's error body)
400	claude-opus-5-5	31	messages.N.output_config: extra inputs not permitted
400	claude-sonnet-5	29	thinking.type.disabled not supported for this model
503	claude-opus-5-5	5	Bedrock unable to process request (503)
400	claude-opus-5-5	2	web_search tool type not supported

Response time of successful (200) requests

n	Mean s	p50 s	p90 s	p99 s	Max s	Sum s
133	18.19	8.75	41.58	166.03	183.57	2,420

Context the agent received

Instruction file injected into the agent's prompts	Requests containing it	of total
~/arjua-forks/CLAUDE.md	196	200
~/arjua-forks/.../2026-10-09_162748/agents/captain-nemo/CLAUDE.md	196	200

Gateway: 200 requests in 29 gateway sessions. The first file is the arjua-forks repository's own project instructions (the earlier throughput-profiling project), loaded because the agent's worktree sits below that directory; the second is the task's CLAUDE.md written by CORAL.

Measurements not available for this run

Quantity	Status
Noop baseline in this run	None evaluated; the comparison point here is the unmodified S3-FIFO seed (attempts #1 and #29).
Per-operation latency (READ p99, SCAN p99), per-thread throughput	Written to a per-eval scratch directory that is deleted after each eval; not retained
Policy counters	Exported only by the agent's own instrumented variants (attempt #13 onward, as policy_counters in the feedback); the seed exports none
Agent notes	0 saved; the agent's attempts to write experiment notes were denied by the permission layer (see permission denials)
CPU utilisation / cycles during evals	Not collected